Exercises 4.4 7, 11, 13, 15, 17, 19, 21, 23, 27, 31, 33, 35, 39, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 79.

11. $\lim_{x\to 1} \frac{x^7-1}{x^3-1}$

$\overset{H}{=} \lim_{x\to 1} \frac{7x^6}{3x^2}$

$= \frac{7}{3}$

13. $\lim_{t\to 0} \frac{e^{2t}-1}{\sin t}$

$\overset{H}{=} \lim_{t\to 0} \frac{2e^{2t}}{\cos t}$

$= 2$

19. $\lim_{x\to\infty} \frac{\sqrt{x}}{1+e^x}$

$\overset{H}{=} \lim_{x\to\infty} \frac{\frac{1}{2\sqrt{x}}}{e^x}$

$= \lim_{x\to\infty} \frac{1}{2\sqrt{x}e^x} = 0$

15. $\lim_{x\to\frac{\pi}{4}} \frac{\sin x - \cos x}{\tan x - 1}$

$\overset{H}{=} \lim_{x\to\frac{\pi}{4}} \frac{\cos x + \sin x}{\sec^2 x}$

$= 2$

17. $\lim_{x\to 1} \frac{\sin(x-1)}{x^3+x-2}$

$\overset{H}{=} \lim_{x\to 1} \frac{\cos(x-1)}{3x^2+1}$

$= \frac{1}{4}$

21. $\lim_{x\to 0^+} \frac{\ln x}{x}$

$= \lim_{x\to 0^+} \frac{-\infty}{0^+}$

$= -\infty$

23. $\lim_{x\to 3} \frac{\ln(\frac{x}{3})}{3-x}$

$\overset{H}{=} \lim_{x\to 3} \frac{\frac{1}{x}}{-1}$

$= \frac{1}{3}$

25

31. $\lim_{x\to 0} \frac{\sin^{-1} x}{x}$

$= 0 \times \overset{H}{\lim_{x\to 0}} \frac{\frac{1}{\sqrt{1-x^2}}}{1}$

$= 1$

35. $\lim_{x\to 0} \frac{\ln(1+x)}{\cos x + e^x - 1}$

$= 0$

45. $\lim_{x\to 0} \sin(5x)\csc 3x$

$= \lim_{x\to 0} \frac{\sin 5x}{\sin 3x}$

$\overset{H}{=} \lim_{x\to 0} \frac{5\cos 5x}{3\cos 3x}$

$= \frac{5}{3}$

47. $\lim_{x\to 0} \frac{e^x + e^{-x} - 2}{e^x - x - 1}$

$\overset{H}{=} \lim_{x\to 0} \frac{e^x - e^{-x}}{e^x - 1}$

$\overset{H}{=} \lim_{x\to 0} \frac{e^x + e^{-x}}{e^x}$

$= 2$

33. $\lim_{x\to 0} \frac{x \cdot 3^x}{3^x - 1}$

$\overset{H}{=} \lim_{x\to 0} \frac{3^x + x \cdot 3^x \ln 3}{3^x \ln 3}$

$= \frac{1}{\ln 3}$

39. $\lim_{x\to 1} \frac{x^a - 1}{x^b - 1}$

$\overset{H}{=} \lim_{x\to 1} \frac{ax^{a-1}}{bx^{b-1}} = \frac{a}{b}$

47. $\lim_{x\to\infty} x^3 e^{-x^2}$

$= \lim_{x\to\infty} \frac{x^3}{e^{x^2}}$

$\overset{H}{=} \lim_{x\to\infty} \frac{3x^2}{e^{x^2} \cdot 2x} = \lim_{x\to\infty} \frac{3x}{2e^{x^2}}$

$\overset{H}{=} \lim_{x\to\infty} \frac{6x}{\quad} \overset{H}{=} \lim_{x\to\infty} \frac{3}{4xe^{x^2}} = 0$

49. $\lim_{x\to 1^+} \ln x \tan\left(\frac{\pi x}{2}\right)$

$= \lim_{x\to 1^+} \frac{\sin\left(\frac{\pi x}{2}\right)\ln x}{\cos\left(\frac{\pi x}{2}\right)}$

$= -\frac{2}{\pi}$

51. $\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{\ln x}\right)$

$= \lim_{x\to 1}\left(\frac{x\ln x - x + 1}{(x-1)\ln x}\right)$

$\overset{H}{=} \lim_{x\to 1} \frac{\ln x}{\ln x + \frac{x-1}{x}} = \lim_{x\to 1} \frac{\ln x}{\ln x - \frac{1}{x} + 1}$

$\overset{H}{=} \lim_{x\to 1} \frac{\frac{1}{x}}{\frac{1}{x} - \frac{1}{x^2}} = \frac{1}{2}$

53. $\lim_{x\to 0^+}\left(\frac{1}{x} - \frac{1}{e^x - 1}\right)$

$= \lim_{x\to 0^+}\left[\frac{e^x - 1 - x}{x(e^x - 1)}\right]$

$\overset{H}{=} \lim_{x\to 0^+} \frac{e^x - 1}{e^x - 1 + xe^x}$

$\overset{H}{=} \lim_{x\to 0^+} \frac{e^x}{e^x + e^x + xe^x} = \frac{1}{2}$

55. $\lim_{x\to 0^+} x^{\sqrt{x}}$

Let $f(x) = x^{\sqrt{x}}$ $\quad \ln f(x) = \sqrt{x}\ln x$

$\lim_{x\to 0^+} \ln f(x) = \lim_{x\to 0^+} \sqrt{x}\ln x$

$= \lim_{x\to 0^+} \frac{\ln x}{x^{-\frac{1}{2}}} \overset{H}{=} \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{2}x^{-\frac{3}{2}}} = \lim_{x\to 0^+} -2\sqrt{x} = 0$

$\lim_{x\to 0^+} x^{\sqrt{x}} = 1$

57. $\lim_{x\to 0}(1-2x)^{\frac{1}{x}}$

Let $y = (1-2x)^{\frac{1}{x}}$

$\ln y = \frac{1}{x}\ln(1-2x)$

$\lim_{x\to 0} \ln y = \lim_{x\to 0} \frac{\ln(1-2x)}{x}$

$\overset{H}{=} \lim_{x\to 0} \frac{\frac{-2}{1-2x}}{1}$

$= -2$

$y = e^{-2}$

59. $\lim_{x\to 1^+} x^{\frac{1}{1-x}}$

Let $y = x^{\frac{1}{1-x}}$

$\ln y = \frac{1}{1-x}\ln x$

$\lim_{x\to 1^+} \ln y = \lim_{x\to 1^+} \frac{\ln x}{1-x} \overset{H}{=} \lim_{x\to 1^+} \frac{\frac{1}{x}}{-1} = -1$

$y = e^{-1}$

66 $\lim_{x \to 0^+} (1 - \cos x)^{\sin x}$

let $y = (1 - \cos x)^{\sin x}$

$\ln y = \sin x \ln(1 - \cos x)$

$\csc x = \frac{1}{\sin x}$

$\sec x = \frac{1}{\cos x}$

$\lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} \sin x \ln(1 - \cos x)$

$= \lim_{x \to 0^+} \frac{\ln(1 - \cos x)}{\csc x}$

$\overset{H}{=} \lim_{x \to 0^+} \frac{\frac{\sin x}{1 - \cos x}}{-\csc x \cot x}$